The Journal of Molecular Diagnostics, Vol. 24, No. 8, August 2022

ELSEVIER

the Journal of Molecular Diagnostics
jmdjournal.org

Validation of Whole Genome Methylation Profiling Classifier for Central Nervous System Tumors

Check for updates

Lucas Santana-Santos,* Kwok Ling Kam,† David Dittmann,* Stephanie De Vito,* Matthew McCord,* Pouya Jamshidi,* Hailie Fowler,‡ Xinkun Wang,‡ Alan M. Aalsburg,‡ Daniel J. Brat,* Craig Horbinski,*§ and Lawrence J. Jennings*

From the Departments of Pathology and Neurological Surgery,§ Feinberg School of Medicine, and the Center for Genomic Medicine,‡ Northwestern University, Chicago, Illinois; and the Department of Pathology,† Beaumont Hospital, Royal Oak, Michigan*

Accepted for publication April 26, 2022.

Address correspondence to Lawrence J. Jennings, M.D, Ph.D., Northwestern University, 710 N. Fairbanks Ct., Olson Pavilion, Rm. 2-513, Chicago, IL 60611. E-mail: l-jennings@northwestern.edu.

The 2021 WHO Classification of Tumors of the Central Nervous System includes several tumor types and subtypes for which the diagnosis is at least partially reliant on utilization of whole genome methylation profiling. The current approach to array DNA methylation profiling utilizes a reference library of tumor DNA methylation data, and a machine learning–based tumor classifier. This approach was pioneered and popularized by the German Cancer Research Network (DKFZ) and University Hospital Heidelberg. This research group has kindly made their classifier for central nervous system tumors freely available as a research tool via a web-based portal. However, their classifier is not maintained in a clinical testing environment. Therefore, the Northwestern Medicine (NM) classifier was developed and validated. The NM classifier was validated using the same training and validation data sets as the DKFZ group. Using the DKFZ validation data set, the NM classifier's performance showed high concordance (92%) and comparable accuracy (specificity 94.0% versus 84.9% for DKFZ, sensitivity 88.6% versus 94.7% for DKFZ). Receiver-operator characteristic curves showed areas under the curve of 0.964 versus 0.966 for NM and DKFZ classifiers, respectively. In addition, in-house validation was performed and performance was compared using both classifiers. The NM classifier performed comparably well and is currently offered for clinical testing. *(J Mol Diagn 2022, 24: 924–934; https://doi.org/10.1016/j.jmoldx.2022.04.009)*

Historically, central nervous system (CNS) tumor classification and grading have been based on histology. Although histology provides crucial diagnostic information, there is inherent subjectivity and interobserver variability in diagnoses based solely on morphology. The importance of advanced molecular testing for diagnosis, prognosis, and targeted therapy has made ancillary molecular studies integral for classification and grading of CNS tumors today.[1] The 2016 and, more recently, the 2021 World Health Organization CNS tumor classification system,[2] list many important genetic alterations for diagnoses (eg, *IDH*, *ATRX*, *TP53*, *TERT* promoter, *H3F3A*, *BRAF*, *MYB*, *MYBL1*, *MN1*, etc.).

Capitalizing on the genomic signature of a tumor, array DNA methylation profiling has proven valuable for the classification of tumors with diagnostically challenging or unusual morphology.

It is also useful for identifying subtypes within certain tumor families.[3,4] Furthermore, it has proven valuable in characterizing new tumor classes.[5] For instance, ependymomas are a morphologically similar family of tumors. However, DNA methylation profiling has defined biologically distinct types, which occur in distinct anatomic locations and distinct patient populations.[6,7] Some ependymoma types are characterized by distinct genetic alterations (ie, *ZFTA1* and *YAP1* fusions). Other types, such as posterior fossa groups A and B, lack distinct mutations and are most reliably distinguished by DNA methylation profiling.[7] Medulloblastoma molecular subgroups were originally defined by transcriptome analysis.[8] They can also be distinguished

Disclosures: None declared.

Copyright © 2022 Association for Molecular Pathology and American Society for Investigative Pathology. Published by Elsevier Inc.
This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0).
https://doi.org/10.1016/j.jmoldx.2022.04.009

by DNA methylation profiling.[9,10] Morphology-based grading of meningiomas is complex, and certain morphologic grading criteria are subject to interobserver variability.[11] DNA methylation profiling has proven to be an objective, reliable tool in predicting risk of recurrence in meningioma.[12,13] Methylation profiling has also proven useful in identifying pleomorphic xanthoastrocytoma and anaplastic pleomorphic xanthoastrocytoma, which can have quite heterogeneous morphology.[14,15] DNA methylation profiling revealed that the tumor class formerly known as primitive neuroectodermal tumors of the CNS, which are characterized by small round blue cell histology, is actually composed of a number of biologically distinct tumor groups with unique genetic drivers.[5]

The current approach to array DNA methylation profiling utilizes a reference library of tumor DNA methylation data, and a machine learning–based tumor classifier. This approach was pioneered and popularized by the German Cancer Research Network (DKFZ) and University Hospital Heidelberg.[3,4] This research group has kindly made their classifier for central nervous system tumors freely available as a research tool via a web-based portal. However, this classifier is not maintained in clinical testing environment. Therefore, the Northwestern Medicine (NM) classifier of CNS tumors was developed and validated. The NM classifier was trained using the same 2801 publicly available samples, as well as the same normalization strategies that were used by the DKFZ group, and validated using the same 1104 publicly available samples. Additional validation data were collected in-house and analyzed using both classifiers for comparison.

Materials and Methods

Data Sets

A previously published data set[3] comprising 3905 central nervous tumor samples was downloaded from GEO (*https://www.ncbi.nlm.nih.gov/geo*; accession number GSE109381). From that data set, 2801 samples were used in the original classifier training, and 1104 were used for validation, respectively labeled reference set and validation set in the GEO series metadata. In-house validation samples are available in GEO under the accession number GSE198855.

Data Extraction and Normalization

Signal intensities were extracted from the methylation array raw data (IDAT files) using the minfi R package version 1.40.0).[16] Raw signals were normalized according to methods previously described.[3] Normalized intensities were used to generate beta values according to Illumina's guidelines (Illumina, San Diego, CA). No batch effects were detected, and therefore, no correction was applied (data not shown).

After normalization, probes in the following categories were removed: i) probes targeting sex chromosomes (n = 11,551); ii) probes containing single nucleotide polymorphisms in targeted CpG or within 5 bp of it (n = 7998); and iii) probes matching to more than one region in the human genome (n = 3965). Publicly available data were analyzed using Illumina Infinium HumanMethylation450 BeadChip (450k), whereas in-house samples were analyzed using Illumina Infinium Human MethylationEPIC BeadChip (EPIC, 850K). Therefore, to make the classifier compatible with both arrays, uncommon probes were removed (n = 33,199). A total of 428,799 probes were used in downstream analysis.

Unsupervised Analysis

Unsupervised clustering was performed using the same methodology as previously described.[3] First, the 32,000 most variable probes from 2801 training samples were selected. Eigenvalue decomposition was performed, and 94 nontrivial components were used for t-Distributed Stochastic Neighbor Embedding (t-SNE) analysis. The RSpectra version 0.16-0 (R Project for Statistical Computing; *https://cran.r-project.org/package=RSpectra*) and Rtsne version 0.15 (R Project for Statistical Computing; *https://cran.r-project.org/web/packages/Rtsne/index.html*) packages were used for principal component analysis and t-SNE analysis, respectively. Following parameters were used for t-SNE analysis: theta = 0, pca = F, max_iter = 2500.

Classifier Development and Score Recalibration

Previous studies showed good classification performance of CNS tumors with a Random Forest classifier followed by score recalibration using an L2-penalized, multinomial logistic regression.[3] For the NM classifier, a similar framework was used and training was performed using the original training data set of 2801 samples. First, the 100,000 topmost variable probes from 428,799 probes were selected. This filter reduced computational requirements without affecting classifier performance. Next, the most important features were selected by training a Random Forest classifier using the filtered 100,000 probes. The most important 10,000 features were selected and utilized in classifier training. Importance was measured by the decrease in class-specific classification accuracy when the feature was permutated. Features were ranked by the minimal rank of the variable importance across all classes. The final classification model was trained by fitting 10,000 trees (mtry parameter = 100) with the 10,000 most important probes selected as previously described.[3] Random forest classifier was trained using the R package randomForest version 4.7-1 (R Project for Statistical Computing; *https://cran.r-project.org/web/packages/randomForest/index.html*).

Score recalibration is necessary to translate random scores to meaningful class probabilities that can be used in a clinical context. Therefore, the L2-penalized, multinomial, logistic regression was fitted, which takes random forest scores as dependent variables, and methylation class as the response variable. To prevent overfitting and bias to unseen cases, independent random forest scores are needed to fit in the logistic regression. Therefore, random forest scores generated by a nested fivefold cross-validation were used. The glmnet package version 4.1-3 (R Project for Statistical Computing; *https://cran.r-project.org/web/packages/glmnet/index.html*) was used to fit this model.

Classifier Validation

During training, the random forest classifier accuracy was assessed using fivefold, nested cross-validation following the workflow previously described.[3] In each cross-validation iteration, one-fifth of the data was used as an independent validation set, whereas the remaining four-fifths was used as a training set. After the fifth iteration, all 2801 training samples would have been predicted by an independently trained predictor.

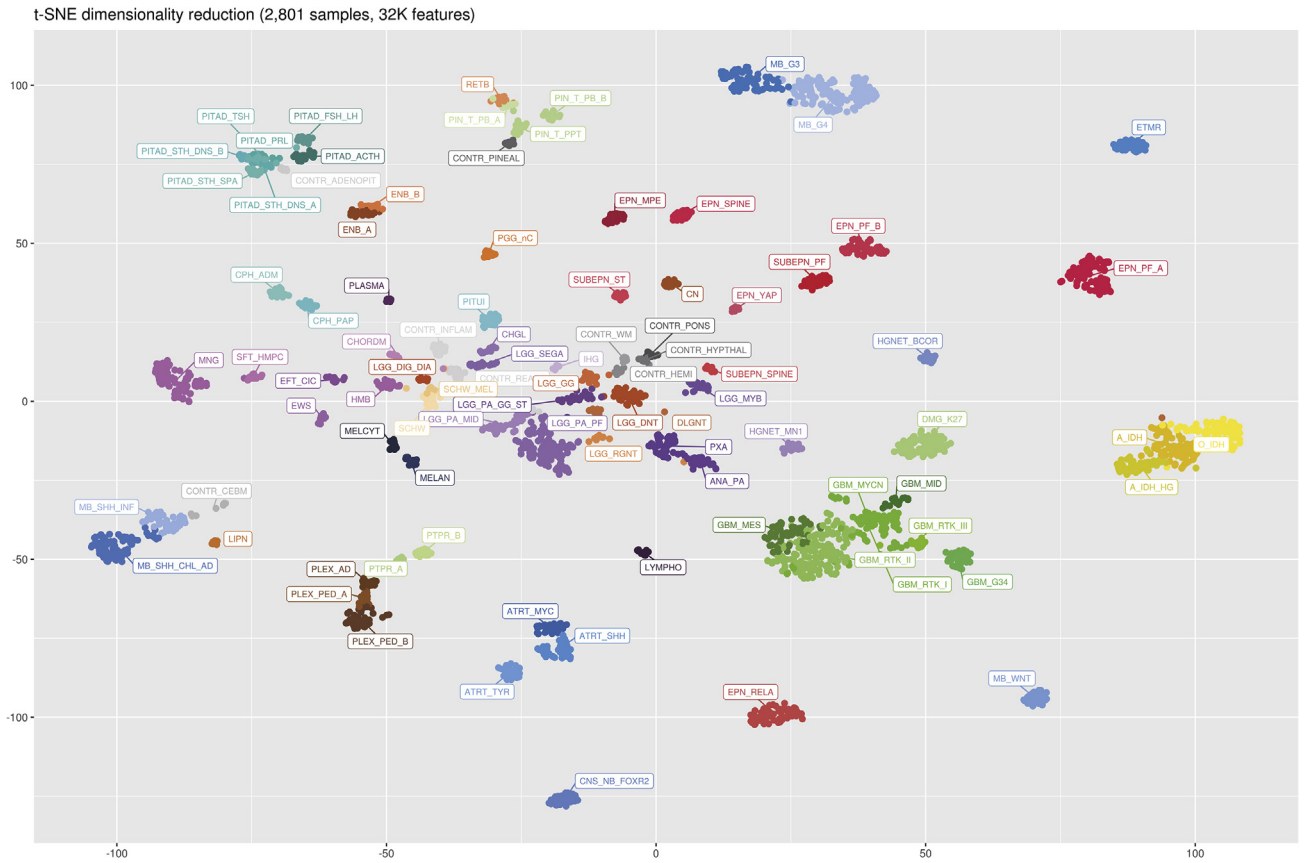
The final random forest classifier and score recalibration logistic regression were also validated using an independent data set of 1104 CNS tumor cases. Additional validation was performed by profiling samples collected in-house and analyzed using both the DKFZ and NM classifiers for comparison.

Samples and DNA Extraction

Samples included in the study were all from human brain sample, which was formalin-fixed for 8 hours followed by paraffin-embedding (FFPE) and stored at room temperature. All samples were extracted using either Covaris (FFPE DNA Kit; Covaris, Inc., Woburn, MA) or Purigen Biosystems (FFPE DNA Kit; Purigen Biosystems, Pleasanton, CA) kits following the manufacturer's instructions. Twenty samples were extracted using both methods as part of the assessment of precision.

Quality Check of FFPE DNAs

The DNA from FFPE blocks was checked for suitability for FFPE restoration, as indicated by the Infinium HD FFPE QC Assay (Illumina, Inc.), by performing a quantitative PCR with 2 ng of FFPE DNA. Delta quantification cycles (ΔCq) was calculated by subtracting the average value of Cq of the interrogated sample from the Cq value of a standard provided by the manufacturer. All FFPE samples had a ΔCq



<5, which is the recommended threshold for suitability for FFPE restoration.

Bisulfite Conversion

Samples ranging from 50 ng to 500 ng of FFPE DNA were pipetted into strip tubes and processed using the EZ-96 DNA Methylation kit (Zymo Research Corp., Tustin, CA) following the manufacturer's recommendations for Infinium assays.

FFPE Restoration

Bisulfite-converted DNA (8 μL) from FFPE samples was processed as previously described.[17] Briefly, the DNA was denatured with 4 μL of NaOH 0.1N for 10 minutes at room temperature. A 1-hour reaction at 37°C was then performed with Primer Pre Restore and Amplification Mix Restore reagents supplied by the kit manufacturer (Illumina, Inc.). DNA was cleaned with a ZR-96 DNA Clean & Concentrator-5 kit (Zymo Research Corp.) and eluted in 13 μL of Elution Restore Buffer reagent (Illumina, Inc.). Cleaned DNA was then denatured for 2 minutes at 95°C, followed by ligation incubation at 37°C for 1 hour with Convert Master Mix reagents (Illumina, Inc.). The resulting material was cleaned with ZR-96 DNA Clean & Concentrator-5 kit (Zymo Research Corp.) and eluted in 10 μL of deionized water.

Array Hybridization

MethylationEPIC beadarray was performed according to the Infinium HD Methylation Assay Protocol, as previously described with minor changes.[16] Eight microliters of restored FFPE bisulfite-converted DNA was used for each sample, and 28 μL of hybridization volume was used when loading the microarray.

Figure 1 Clustering of 2801 reference central nervous system (CNS) tumor samples using t-Distributed Stochastic Neighbor Embedding (t-SNE) dimensionality reduction. Only the top 32,000 most variable probes across samples were utilized for t-SNE analysis. Methylation class labels point to the corresponding group centroid coordinates. Methylation classes are color-coded according to sample histology. Control tissues are represented in gray; hematopoietic tumors in dark purple; nerve tumors in beige; mesenchymal tumors in pink; sellar tumors in cyan; melanocytic tumors in dark blue; ependymomas in red; glioneuronal tumors in orange; isocitrate dehydrogenase (IDH)-mutated gliomas in yellow; choroid plexus tumors in brown; pineal region tumors in mint green; glioblastomas in green; and other gliomas in violet. Abbreviations represent the following methylation groups: A_IDH, astrocytoma; A_IDH_HG, high grade astrocytoma; ANA_PA, anaplastic pilocytic astrocytoma; ATRT_MYC, MYC; ATRT_SHH, SHH; ATRT_TYR, TYR; CHGL, methylation class chordoid glioma of the third ventricle; CHORDM, methylation class chordoma; CN, methylation class central neurocytoma; CNS_NB_FOXR2, methylation class CNS neuroblastoma with FOXR2 activation; CONTR_ADENOPIT, methylation class control tissue, pituitary gland anterior lobe; CONTR_CEBM, methylation class control tissue, cerebellar hemisphere; CONTR_HEMI, methylation class control tissue, hemispheric cortex; CONTR_HYPTHAL, methylation class control tissue, hypothalamus; CONTR_INFLAM, methylation class control tissue, inflammatory tumor microenvironment; CONTR_PINEAL, methylation class control tissue, pineal gland; CONTR_PONS, methylation class control tissue, pons; CONTR_REACT, methylation class control tissue, reactive tumor microenvironment; CONTR_WM, methylation class control tissue, white matter; CPH_ADM, methylation class craniopharyngioma, adamantinomatous; CPH_PAP, methylation class craniopharyngioma, papillary; DLGNT, methylation class diffuse leptomeningeal glioneuronal tumor; DMG_K27, methylation class diffuse midline glioma H3 K27M mutant; EFT_CIC, methylation class CNS Ewing sarcoma family tumor with CIC alteration; ENB_A, methylation class esthesioneuroblastoma, subclass A; ENB_B, methylation class esthesioneuroblastoma, subclass B; EPN_MPE, methylation class ependymoma, myxopapillary; EPN_PF_A, methylation class ependymoma, posterior fossa group A; EPN_PF_B, methylation class ependymoma, posterior fossa group B; EPN_RELA, methylation class ependymoma, RELA fusion; EPN_SPINE, methylation class ependymoma, spinal; EPN_YAP, methylation class ependymoma, YAP fusion; ETMR, methylation class embryonal tumor with multilayered rosettes; EWS, methylation class Ewing sarcoma; GBM_G34, methylation class glioblastoma, IDH wildtype, H3.3 G34 mutant; GBM_MES, mesenchymal; GBM_MID, midline; GBM_MYCN, MYCN; GBM_RTK_I, RTK I; GBM_RTK_II, RTK II; GBM_RTK_III, RTK III; HGNET_BCOR, methylation class CNS high grade neuroepithelial tumor with BCOR alteration; HGNET_MN1, methylation class CNS high grade neuroepithelial tumor with MN1 alteration; HMB, methylation class hemangioblastoma; IHG, methylation class infantile hemispheric glioma; LGG_DIG_DIA, methylation class low grade glioma, desmoplastic infantile astrocytoma/ganglioglioma; LGG_DNT, methylation class low grade glioma, dysembryoplastic neuroepithelial tumor; LGG_GG, methylation class low grade glioma, ganglioglioma; LGG_MYB, methylation class low grade glioma, MYB/MYBL1; LGG_PA_GG_ST, methylation class low grade glioma, subclass hemispheric pilocytic astrocytoma and ganglioglioma; LGG_PA_MID, midline pilocytic astrocytoma; LGG_PA_PF, posterior fossa pilocytic astrocytoma; LGG_RGNT, methylation class low grade glioma, rosette forming glioneuronal tumor; LGG_SEGA, methylation class low grade glioma, subependymal giant cell astrocytoma; LIPN, methylation class cerebellar liponeurocytoma; LYMPHO, methylation class lymphoma; MB_G3, group 3; MB_G4, group 4; MB_SHH_CHL_AD, SHH A (children and adult); MB_SHH_INF, SHH B (infant); MB_WNT, WNT; MELAN, methylation class melanoma; MELCYT, methylation class melanocytoma; MNG, methylation class meningioma; O_IDH, 1p/19q codeleted oligodendroglioma; PGG_nC, methylation class paraganglioma, spinal non-CIMP; PIN_T_PB_A, methylation class pineoblastoma group A/intracranial retinoblastoma; PIN_T_PB_B, methylation class pineoblastoma group B; PIN_T_PPT, methylation class pineal parenchymal tumor; PITAD_ACTH, methylation class pituitary adenoma, ACTH; PITAD_FSH_LH, methylation class pituitary adenoma, FSH/LH; PITAD_PRL, methylation class pituitary adenoma, prolactin; PITAD_STH_DNS_A, methylation class pituitary adenoma, STH densely granulated, group A; PITAD_STH_DNS_B, methylation class pituitary adenoma, STH densely granulated, group B; PITAD_STH_SPA, methylation class pituitary adenoma, STH sparsely granulated; PITAD_TSH, methylation class pituitary adenoma, TSH; PITUI, methylation class pituicytoma/granular cell tumor/spindle cell oncocytoma; PLASMA, methylation class plasmacytoma; PLEX_AD, adult; PLEX_PED_A, paediatric A; PLEX_PED_B, paediatric B; PTPR_A, methylation class papillary tumor of the pineal region group A; PTPR_B, methylation class papillary tumor of the pineal region group B; PXA, methylation class (anaplastic) pleomorphic xanthoastrocytoma; RETB, methylation class retinoblastoma; SCHW, methylation class schwannoma; SCHW_MEL, methylation class melanotic schwannoma; SFT_HMPC, methylation class solitary fibrous tumor/hemangiopericytoma; SUBEPN_PF, methylation class subependymoma, posterior fossa; SUBEPN_SPINE, methylation class subependymoma, spinal; SUBEPN_ST, methylation class subependymoma, supratentorial.

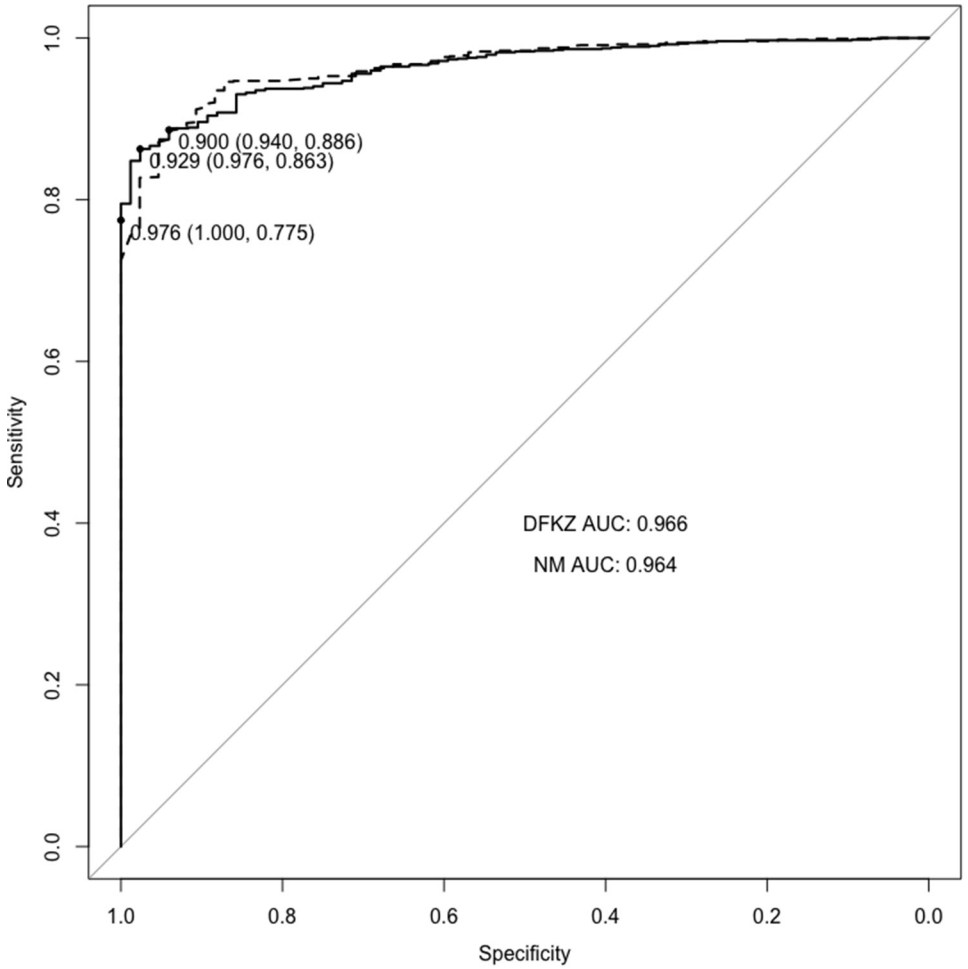


Figure 2 Receiver-operating characteristic (ROC) curve for Northwestern Medicine (NM) and German Cancer Research Network (DKFZ) classifiers using 1104 samples. Curve corresponding to the NM classifier is represented by a **solid black line**, whereas the curve for the DKFZ classifier is represented by a **dashed black line**. Area under the curve (AUC) values are 0.966 and 0.964 for DKFZ and NM classifiers, respectively. Specificity of 1 and a sensitivity of 0.775 is achieved when a score of 0.976 is used. Specificity of 0.940 and a sensitivity of 0.886 is achieved when a score of 0.9 is used. Younden index corresponds to a score of 0.929 with a specificity of 0.976 and sensitivity of 0.863.

Results

Accuracy of the NM Classifier

To establish a comprehensive CNS tumor reference cohort, genome-wide DNA methylation profiles of 2801 samples were used as previously described.[3] Unsupervised clustering was performed using t-SNE (Figure 1), which shows the presence of 91 distinct methylation classes in agreement with previous observations.[3]

A classifier was developed using the Random Forest algorithm as previously described.[3] Class probability is determined using a fitted L2-penalized, multinomial, logistic regression model that transforms the raw score into a probability or calibrated score. To assess the validity of this classifier, an independent group of 1104 samples was analyzed.[3] The area under the curve was calculated for both the DKFZ and NM classifiers (Figure 2). Both classifiers showed comparable areas under the curve: 0.966 and 0.964 for DKFZ and NM classifiers, respectively (Figure 2). A score of 0.976 achieves a specificity of 1 and sensitivity of 0.775, whereas the Youden index corresponds to a score of 0.929 with a specificity of 0.976 and sensitivity of 0.863. A score of 0.9 achieves a better balance between specificity and sensitivity (0.940 and 0.886, respectively) than the previous two thresholds, and will be used for subsequent analysis (Figure 2). In comparison, the DKFZ classifier achieved a specificity of 0.849 and a sensitivity of 0.947 when the recommended score of 0.9 was applied.

Concordance of the 1104 samples as compared to the DKFZ classifier can be found in Figure 3. Inaccurately classified samples are shown, and some can be seen to have high calibrated scores using one classifier or the other, but not both, demonstrating the potential to reduce misclassification through the utilization of more than one classifier.

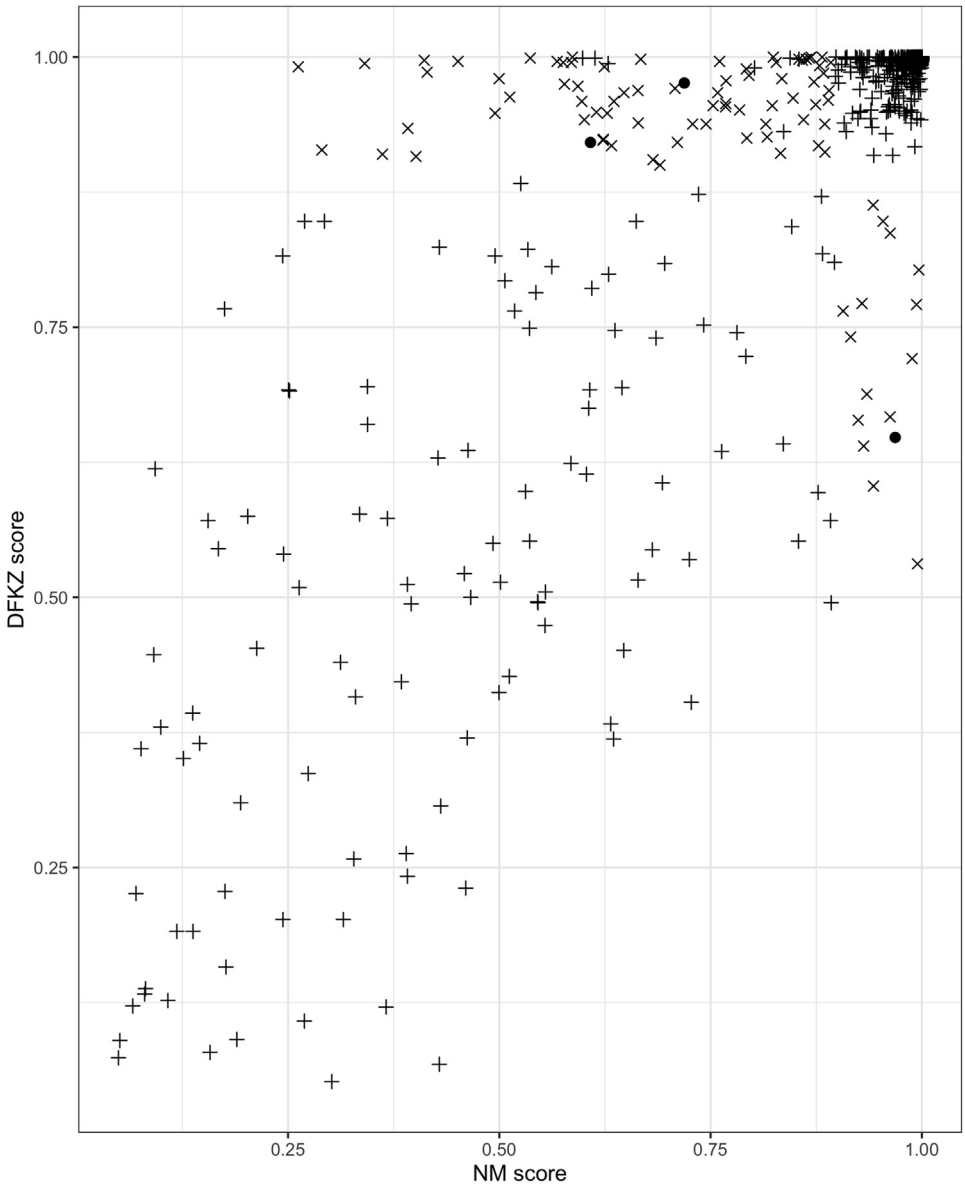


Figure 3 Scatterplot of Northwestern Medicine (NM) and German Cancer Research Network (DKFZ) scores for 1104 samples. **Crosses** show samples that are concordant between NM and DKFZ classifiers. Samples with discordant scores, but concordant classifications, are represented by **Xs**. Discordant samples are represented by **circles**.

DKFZ and NM concordance was 92% (1012/1104) when a cutoff of 0.9 was applied for both classifiers (Supplemental Table S1). Out of the 92 nonconcordant cases, 88 cases had the same classification, only differing in the magnitude of the scores between classifiers. Classifiers also showed similar score distributions, as NM scores median was 0.997 with a SD of 0.197, and DKFZ median score was 0.999 with a SD of 0.167.

Accuracy of In-House Samples

To assess the overall accuracy of the methylation array, genome-wide DNA methylation profiles were performed on 53 unique samples across a range of tumor types (Supplemental Table S2). The diagnoses were determined using the combination of histology, location, and history together with ancillary studies when available (eg, immunohistochemistry, next-generation sequencing, single nucleotide polymorphism microarray, and methylation array). Accuracy, when compared with integrated sample diagnosis, was 69.8% (37/53) when a cutoff of 0.9 was applied, and 90.5% (48/53) was achieved when the classification was considered without a score threshold (Supplemental Table S2). In comparison, DKFZ accuracy was 77.5% (38/49) for in-house samples when a cutoff of 0.9 was applied, and 91.8% (45/49) was achieved when the classification was considered without a score threshold. It is important to notice that four samples failed

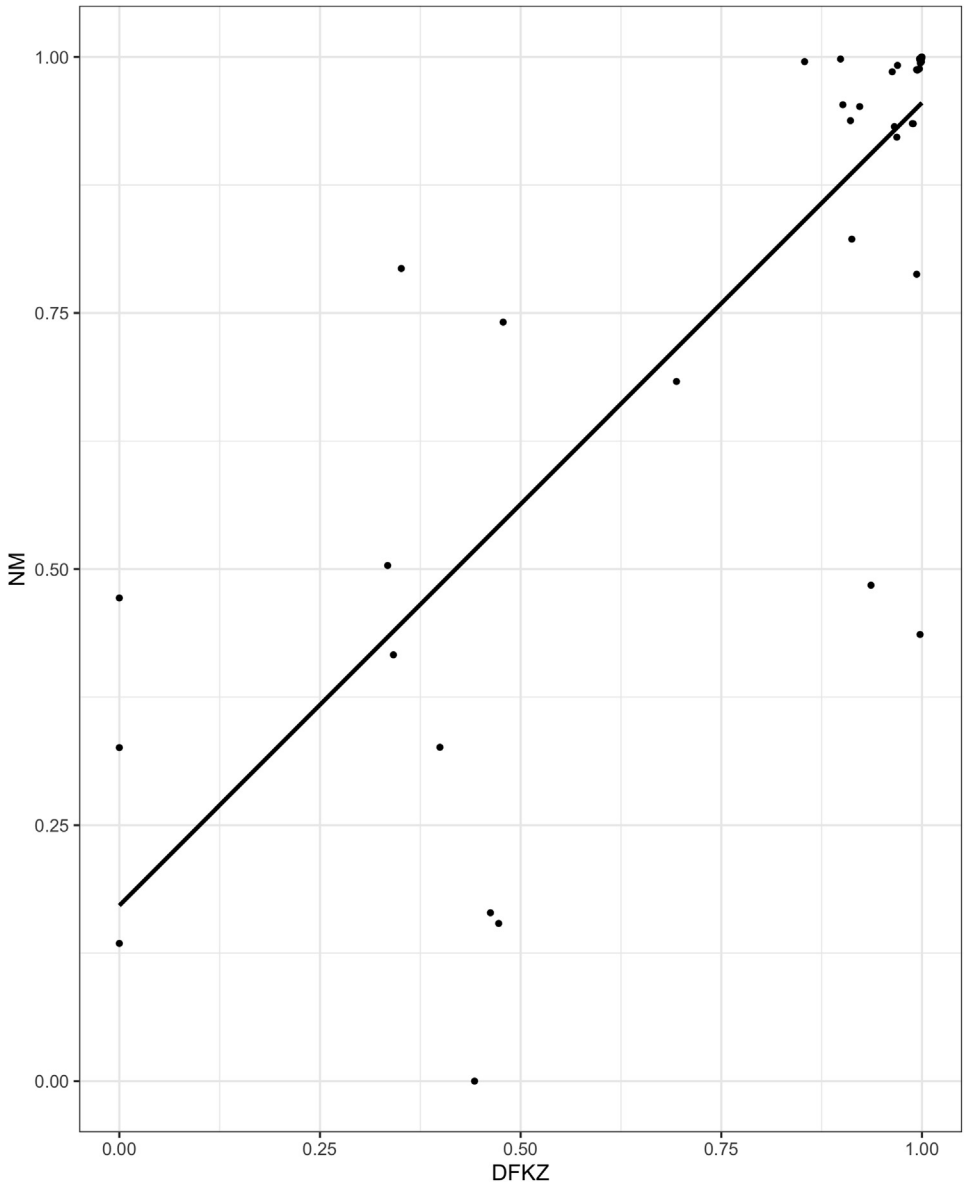


Figure 4 Correlation plot of German Cancer Research Network (DKFZ) and Northwestern Medicine (NM) classification scores for 52 samples. Correlation line is shown ($R^2 = 0.72$).

analysis on the DKFZ website, and two of these samples did not match using the NM classifier, hence contributing to the lower accuracy numbers on the latter. The score correlation between NM and DKFZ was 0.81 with the highest variation among those with lowest scores (<0.5) as seen in Figure 4.

Precision

Samples that were run more than once are shown in Table 1. The following variables were changed to ensure that sample processing was reproducible: date, extraction methods, and technologist. Twenty-four samples were run two or more times. Classification between the sets of samples was concordant for 100% (24/24) of the replicates. There was no appreciable difference in performance for any of the parameters that were assessed. Paired *t*-test showed no statistically significant difference in the scores obtained by samples extracted with the Purigen and Covaris kits ($n = 10$, $P = 0.5911$). Likewise, paired *t*-test showed no statistically significant difference in the scores obtained by samples processed by technologist 1 versus technologist 2 ($n = 7$, $P = 1.00$).

Lower Limit of Detection

The optimal FFPE DNA input is 250 to 500 ng. However, decreasing the sample input to as low as 50 ng had no impact on the accuracy of the classifier (Table 2).

Table 1 Precision Sample Results

Sample set	Extraction method	Technician	Run date	Class	Score	Status
1	Purigen	1	July 23, 2021	Craniopharyngioma, papillary	0.15	No match
	Purigen	1	July 23, 2021		0.18	No match
2	Purigen	1	July 23, 2021	Meningioma	1.00	Match
	Purigen	1	July 7, 2021		1.00	Match
3	Purigen	1	July 23, 2021	Glioblastoma, IDH wildtype, H3.3 G34 mutant	0.98	Match
	Purigen	1	July 7, 2021		0.99	Match
4	Purigen	1	July 30, 2021	No match	0.00	No match
	Purigen	1	July 23, 2021	Control tissue, inflammatory tumor microenvironment	0.11	No match
5	Purigen	1	July 30, 2021	Control tissue, reactive tumor microenvironment	1.00	Match
	Purigen	1	September 7, 2021		0.98	Match
6	Purigen	1	July 30, 2021	IDH glioma	1.00	Match
	Purigen	1	September 7, 2021		1.00	Match
7	Purigen	1	July 23, 2021	Ependymoma, myxopapillary	1.00	Match
	Purigen	1	September 7, 2021		1.00	Match
8	Purigen	2	July 23, 2021	Infantile hemispheric glioma	0.50	No match
	Purigen	1	July 23, 2021		0.48	No match
9	Purigen	1	July 23, 2021	Craniopharyngioma, adamantinomatous	1.00	Match
	Covaris	1	July 30, 2021		1.00	Match
10	Purigen	2	July 23, 2021	Pineal parenchymal tumor	1.00	Match
	Purigen	2	July 23, 2021		1.00	Match
	Purigen	1	July 23, 2021		1.00	Match
11	Purigen	2	July 23, 2021	IDH glioma	1.00	Match
	Purigen	2	July 23, 2021		1.00	Match
12	Purigen	2	July 23, 2021	Schwannoma	0.95	Match
	Covaris	1	July 30, 2021		0.96	Match
	Purigen	1	September 7, 2021		0.94	Match
13	Purigen	2	July 23, 2021	Solitary fibrous tumor/hemangiopericytoma	1.00	Match
	Covaris	1	July 30, 2021		1.00	Match
	Purigen	1	September 7, 2021		1.00	Match
14	Purigen	2	July 23, 2021	Pituitary adenoma, FSH/LH	0.99	Match
	Purigen	2	July 23, 2021		0.99	Match
15	Purigen	2	July 16, 2021	Plexus tumor	1.00	Match
	Purigen	2	July 23, 2021		1.00	Match
16	Purigen	2	July 16, 2021	IDH glioma	1.00	Match
	Covaris	2	July 16, 2021		1.00	Match
17	Purigen	2	July 16, 2021	Meningioma	0.82	No match
	Covaris	2	July 16, 2021		0.82	No match
18	Purigen	2	July 16, 2021	Glioblastoma, IDH wildtype	1.00	Match
	Covaris	2	July 16, 2021		0.99	Match
19	Purigen	2	July 16, 2021	Hemangioblastoma	1.00	Match
	Covaris	2	July 16, 2021		1.00	Match
	Purigen	1	September 7, 2021		1.00	Match
20	Purigen	2	July 16, 2021	Meningioma	1.00	Match
	Covaris	2	July 16, 2021		1.00	Match
21	Covaris	1	July 30, 2021	Medulloblastoma, SHH	0.95	Match
	Purigen	2	July 23, 2021		0.94	Match
	Purigen	2	July 23, 2021		0.93	Match
22	Purigen	2	July 23, 2021	Low-grade glioma, dysembryoplastic neuroepithelial tumor	1.00	Match
	Purigen	2	July 23, 2021		1.00	Match
23	Purigen	2	July 23, 2021	Pilocytic astrocytoma	1.00	Match
	Purigen	2	July 23, 2021		1.00	Match
24	Covaris	1	July 30, 2021	Medulloblastoma group 3 and 4	1.00	Match
	Purigen	2	July 23, 2021		1.00	Match
	Purigen	1	July 23, 2021		1.00	Match

Sample with scores <0.9 are assigned a no match status.

FSH, follicle-stimulating hormone; IDH, isocitrate dehydrogenase; LH, luteinizing hormone; SHH, sonic hedgehog.

Table 2 LOD Results

DNA input, ng	Class	Score
500	Methylation class solitary fibrous tumor/hemangiopericytoma	0.960
250	Methylation class solitary fibrous tumor/hemangiopericytoma	0.962
125	Methylation class solitary fibrous tumor/hemangiopericytoma	0.961
100	Methylation class solitary fibrous tumor/hemangiopericytoma	0.957
50	Methylation class solitary fibrous tumor/hemangiopericytoma	0.961

Classification score and class are not altered by reducing DNA input (nanograms).
LOD, limit of detection.

Interfering Substances

Different possible interfering substances, such as necrosis, blood clots, and tumor cellularity could potentially affect the classification of tumors. Therefore, these were estimated for 44 in-house accuracy samples. No strong correlations were found between interfering substances and classification scores (Figure 5). R^2 values for blood and classification scores, necrosis and classification scores, and tumor cellularity and classification scores were 0.18, 0.25, and 0.03, respectively.

Reportable Range

The calibrated score corresponds to a probability of a match. Scores that fall below the 0.9 threshold for interpretation of a match may still prove informative when considered together with other diagnostic and clinical findings. Therefore, methylation classes and subclasses are reported when the score exceeds 0.1, and scores >0.9 are considered a match for class, whereas scores >0.5 are considered a match for subclass.

Reference Range

There is no reference range for this brain tumor classifier. Although nontumor samples were included in the development and validation of the classifier (eg, pituitary gland anterior lobe, cerebellar hemisphere, hemispheric cortex, hypothalamus, pineal gland, white matter), these diagnostic categories are not considered to be the normal or reference range.

Discussion

The greatest challenge of validating whole genome methylation profiling of brain tumors for clinical testing was the development and validation of the NM classifier. Fortunately, the extensive work of other investigators was made publicly available.[3] Using the same data sets and normalization methods, the NM classifier could be developed and validated with an independent set of data files and showed the NM classifier to be comparably accurate. Interestingly, some miscalled samples can be seen with high confidence by each classifier, but not with both. This variation is attributed to feature selection in the development of the classifier, and shows the potential advantage of having more than one classifier to assess a given sample.

When applied to in-house validation samples and compared with the integrated sample diagnosis, the NM classifier accuracy was 69.8% (37/53) when a cutoff of 0.9 was applied, and 90.5% (48/53) was achieved when the classification was considered without a score threshold versus 77.5% (38/49) and 91.8% (45/49) for DKFZ. It should be noted that the denominator for the DKFZ was 49 because four samples failed analysis on the DKFZ website, and two of these samples did not match using the NM classifier, hence contributing to the lower accuracy numbers on the latter. The accuracy without a threshold closely matched the accuracy of both classifiers (92%) using the 1104 sample data set suggesting that the apparently lower accuracy for the in-house samples could be attributable to lower confidence scores. The lower confidence scores are likely due samples used in the validation, which were older stored DNA samples. Therefore, when confidence scores are low, the quality of the sample should be considered including low tumor cellularity. A chromosomal copy number plot can be very useful to assess the quality and cellularity of the DNA for a particular sample. Poorer quality DNA will have more variance or noise in the copy number estimates and lower tumor cellularity will have smaller shifts. If the quality is deemed good, other reasons for a low confidence score include a new or rare tumor type that was not represented in the classifier or a limitation of the classifier itself. Correlation with clinical history, histology, and ancillary studies is therefore essential for those cases with lower calibrated scores. The t-SNE plot can also be particularly useful for samples with lower confidence scores. The t-SNE plot is a dimension-reduction approach for visualization of high-dimensional data. The data being visualized are unsupervised clustering of all 2801 training samples, and it is a different clustering algorithm than that used to create the tumor classifier, which is done by random forest. For high score samples (score >0.9), t-SNE plots closely align with the classifier results. When the score is <0.9, the t-SNE plot may show proximity to a particular group and provides additional information together with copy number variations, mutations, histology,

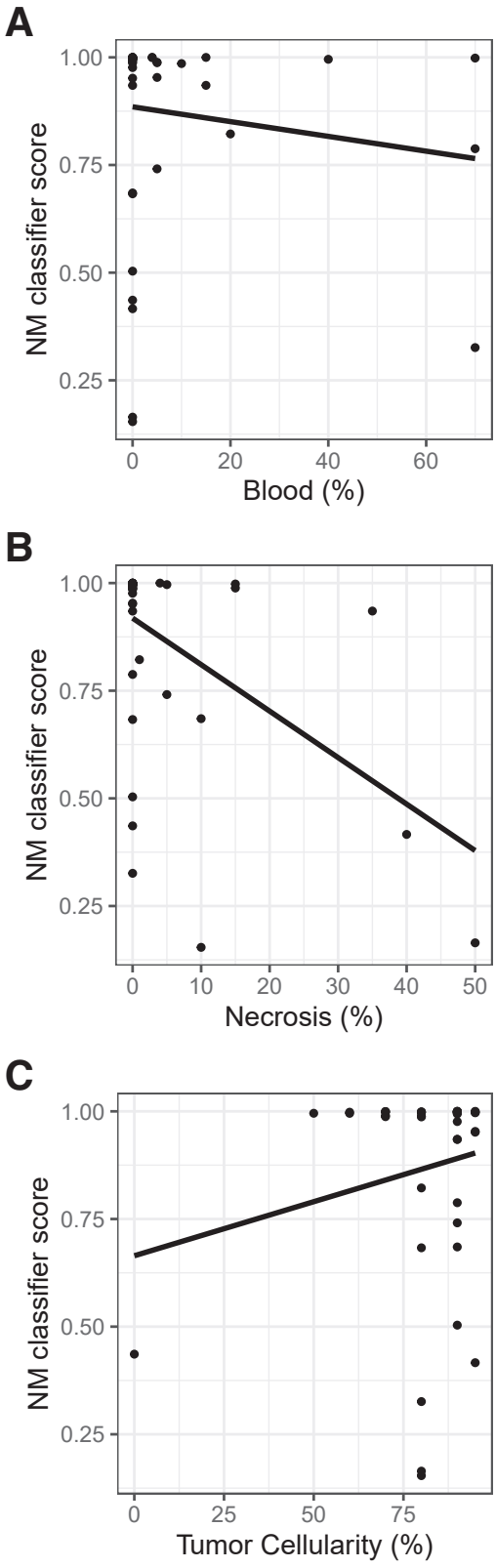


Figure 5 Scatterplots of interfering substances (blood, necrosis, and tumor cellularity) and methylation classifier scores for 44 accuracy samples. Solid lines represent regression lines. **A:** Scatterplot of blood (%) and classifier scores (R^2 = 0.18). **B:** Scatterplot of necrosis and classifier scores (R^2 = 0.25). **C:** Scatterplot of tumor cellularity (%) and classifier scores (R^2 = 0.03).

immunohistochemistry, and clinical presentation to help ascertain the most likely scenario (eg, low tumor cellularity or a novel subtype).

These validation studies show methylation array on FFPE tissue samples to be highly robust and reproducible. Sample-specific issues, such as blood clot, necrosis, and estimated tumor cellularity had no significant effect on calibrated score. Also, repeat testing of 24 samples showed no difference between different extraction methods, technologists or runs and sample input as low as 50 ng showed good performance.

As demonstrated previously,[4–10] whole genome methylation profiling of brain tumors can be especially useful for pediatric brain tumors, unconventional tumors with atypical morphology and immunohistochemistry profile, subtyping of group 3 and 4 medulloblastomas, and subgrouping of ependymoma/pleomorphic xanthoastrocytoma. In addition, it could be used for other diagnostically challenging cases and discovery of new entities that may have therapeutic and prognostic implications. Moreover, other clinically useful information can be obtained through a methylation array, including *MGMT* methylation status and copy number changes. Indeed, others[18] have shown accurate classification of MGMT methylation status using 2 positions on the methylation array. This approach was compared with MGMT methylation status by pyrosequencing as a separate validation and performed very well (data not shown). For copy number changes, as compared with single nucleotide polymorphism array, there are some inherent limitations, including an inability to call loss of heterozygosity or tetraploidy. However, large copy number changes can be easily discerned, and validation of this approach to identify clinically relevant copy number variations in brain tumors is in process.

In summary, whole-genome methylation profiling of brain tumors for clinical testing has been developed and validated. A crucial part of this validation was the development and validation of a classifier that can be adapted and controlled within a Clinical Laboratory Improvement Amendments–certified clinical laboratory setting. Future developments will include updating the classifier to accommodate additional data and clinical entities. Indeed, sharing of deidentified data sets across institutions will be important for the continuing evolution of this exciting area of clinical testing; such endeavors are currently underway.

Author Contributions

D.J.B., C.H., and L.J.J. secured funding and formulated the validation plan; K.L.K., M.M., and P.J. performed case review and selection; K.L.K., D.D., S.D.V., H.F., X.W., and A.M.A. performed testing and analyzed primary data; L.S.-S. developed classifier and performed statistical analysis; L.S.-S., K.L.K., S.D.V., P.J., and L.J.J. wrote the manuscript; L.S.-S., D.J.B., C.H., and L.J.J. reviewed and

edited the final draft. L.J.J. is the guarantor of this work and, as such, had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Supplemental Data

Supplemental material for this article can be found at *http://doi.org/10.1016/j.jmoldx.2022.04.009*.

References

1. Horbinski C, Ligon KL, Brastianos P, Huse JT, Venere M, Chang S, Buckner J, Cloughesy T, Jenkins RB, Giannini C, Stupp R, Nabors LB, Wen PY, Aldape KJ, Lukas RV, Galanis E, Eberhart CG, Brat DJ, Sarkaria JN: The medical necessity of advanced molecular testing in the diagnosis and treatment of brain tumor patients. Neuro Oncol 2019, 21:1498−1508
2. Louis DN, Perry A, Wesseling P, Brat DJ, Cree IA, Figarella-Branger D, Hawkins C, Ng HK, Pfister SM, Reifenberger G, Soffietti R, von Deimling A, Ellison DW: The 2021 WHO Classification of Tumors of the Central Nervous System: a summary. Neuro Oncol 2021, 23:1231−1251
3. Capper D, Jones DTW, Sill M, Hovestadt V, Schrimpf D, Sturm D, et al: DNA methylation-based classification of central nervous system tumours. Nature 2018, 555:469−474
4. Capper D, Stichel D, Sahm F, Jones DTW, Schrimpf D, Sill M, Schmid S, Hovestadt V, Reuss DE, Koelsche C, Reinhardt A, Wefers AK, Huang K, Sievers P, Ebrahimi A, Schöler A, Teichmann D, Koch A, Hänggi D, Unterberg A, Platten M, Wick W, Witt O, Milde T, Korshunov A, Pfister SM, von Deimling A: Practical implementation of DNA methylation and copy-number-based CNS tumor diagnostics: the Heidelberg experience. Acta Neuropathol 2018, 136:181−210
5. Sturm D, Orr BA, Toprak UH, Hovestadt V, Jones DTW, Capper D, et al: New brain tumor entities emerge from molecular classification of CNS-PNETs. Cell 2016, 164:1060−1072
6. Witt H, Gramatzki D, Hentschel B, Pajtler KW, Felsberg J, Schackert G, Löffler M, Capper D, Sahm F, Sill M, von Deimling A, Kool M, Herrlinger U, Westphal M, Pietsch T, Reifenberger G, Pfister SM, Tonn JC, Weller M, German Glioma Network: DNA methylation-based classification of ependymomas in adulthood: implications for diagnosis and treatment. Neuro Oncol 2018, 20:1616−1624
7. Ellison DW, Aldape KD, Capper D, Fouladi M, Gilbert MR, Gilbertson RJ, Hawkins C, Merchant TE, Pajtler K, Venneti S, Louis DN: cIMPACT-NOW update 7: advancing the molecular classification of ependymal tumors. Brain Pathol 2020, 30:863−866
8. Taylor MD, Northcott PA, Korshunov A, Remke M, Cho Y-J, Clifford SC, Eberhart CG, Parsons DW, Rutkowski S, Gajjar A, Ellison DW, Lichter P, Gilbertson RJ, Pomeroy SL, Kool M, Pfister SM: Molecular subgroups of medulloblastoma: the current consensus. Acta Neuropathol 2012, 123:465−472
9. Schwalbe EC, Williamson D, Lindsey JC, Hamilton D, Ryan SL, Megahed H, Garami M, Hauser P, Dembowska-Baginska B, Perek D, Northcott PA, Taylor MD, Taylor RE, Ellison DW, Bailey S, Clifford SC: DNA methylation profiling of medulloblastoma allows robust subclassification and improved outcome prediction using formalin-fixed biopsies. Acta Neuropathol 2013, 125:359−371
10. Korshunov A, Sahm F, Zheludkova O, Golanov A, Stichel D, Schrimpf D, Ryzhova M, Potapov A, Habel A, Meyer J, Lichter P, Jones DTW, von Deimling A, Pfister SM, Kool M: DNA methylation profiling is a method of choice for molecular verification of pediatric WNT-activated medulloblastomas. Neuro Oncol 2019, 21:214−221
11. Rogers CL, Perry A, Pugh S, Vogelbaum MA, Brachman D, McMillan W, Jenrette J, Barani I, Shrieve D, Sloan A, Bovi J, Kwok Y, Burri SH, Chao ST, Spalding AC, Anscher MS, Bloom B, Mehta M: Pathology concordance levels for meningioma classification and grading in NRG Oncology RTOG Trial 0539. Neuro Oncol 2016, 18:565−574
12. Nassiri F, Mamatjan Y, Suppiah S, Badhiwala JH, Mansouri S, Karimi S, Saarela O, Poisson L, Gepfner-Tuma I, Schittenhelm J, Ng H-K, Noushmehr H, Harter P, Baumgarten P, Weller M, Preusser M, Herold-Mende C, Tatagiba M, Tabatabai G, Sahm F, von Deimling A, International Consortium on Meningiomas, Zadeh G, Aldape KD: DNA methylation profiling to predict recurrence risk in meningioma: development and validation of a nomogram to optimize clinical management. Neuro Oncol 2019, 21:901−910
13. Maas SLN, Stichel D, Hielscher T, Sievers P, Berghoff AS, Schrimpf D, et al: Integrated molecular-morphologic meningioma classification: a multicenter retrospective analysis, retrospectively and prospectively validated. J Clin Oncol 2021, 39:3839−3852
14. Vaubel R, Zschernack V, Tran QT, Jenkins S, Caron A, Milosevic D, Smadbeck J, Vasmatzis G, Kandels D, Gnekow A, Kramm C, Jenkins R, Kipp BR, Rodriguez FJ, Orr BA, Pietsch T, Giannini C: Biology and grading of pleomorphic xanthoastrocytoma-what have we learned about it? Brain Pathol 2021, 31:20−32
15. Kam K-L, Snuderl M, Khan O, Wolinsky J-P, Gondi V, Grimm S, Horbinski C: Using methylation profiling to diagnose systemic metastases of pleomorphic xanthoastrocytoma. Neurooncol Adv 2020, 2: vdz057
16. Aryee MJ, Jaffe AE, Corrada-Bravo H, Ladd-Acosta C, Feinberg AP, Hansen KD, Irizarry RA: Minfi: a flexible and comprehensive Bioconductor package for the analysis of Infinium DNA methylation microarrays. Bioinformatics 2014, 30:1363−1369
17. Moran S, Vizoso M, Martinez-Cardús A, Gomez A, Matías-Guiu X, Chiavenna SM, Fernandez AG, Esteller M: Validation of DNA methylation profiling in formalin-fixed paraffin-embedded samples using the Infinium HumanMethylation450 Microarray. Epigenetics 2014, 9:829−833
18. Bady P, Delorenzi M, Hegi ME: Sensitivity analysis of the MGMT-STP27 model and impact of genetic and epigenetic context to predict the MGMT methylation status in gliomas and other tumors. J Mol Diagn 2016, 18:350−361